04/29/2026

Microlocal characterization

of the image of K_I

What we've done so far. For a fan (N,Σ):

(1) Constructed a lax sym monoidal colimit-preserving functor

$$\kappa_\Sigma = R_\Sigma \circ \Psi_\Sigma \circ \Phi_\Sigma^{-1} : (QCoh(X_\Sigma/T), \otimes) \longrightarrow (Sh(M_{\mathbb{R}}), *)$$

- Φ_Σ^{-1} is an equivalence and sym monoidal
- R_Σ is fully faithful and lax sym monoidal
- Ψ_Σ is symmetric monoidal

(2) κ_Σ factors through sheaves constructible for the FLTZ stratification

(3) If Σ is smooth, we showed that Ψ_Σ is fully faithful, so κ_Σ is fully faithful.

(4) If Σ is projective, we showed that R_Σ is an equivalence, so κ_Σ is symmetric monoidal

Today. Characterizing $\mathrm{im}(\kappa_\Sigma)$ when Σ is smooth projective

Remark. FLTZ and Kuwagaki claim the equivalence between $QCoh(X_\Sigma/T)$ and $im(\kappa_\Sigma)$ when Σ is only projective. This result should be correct:

- Bai and Hu's proof of the characterization of $im(\kappa_\Sigma)$ only uses that Σ is projective. So as long as one knows κ_Σ is fully faithful for general Σ, the claim follows.

- One should be able to deduce full faithfulness for general Σ by **toric resolution of singularities** (which is completely combinatorial). By the functoriality of κ_Σ, this amounts to showing that if a fan Σ' subdivides Σ, then the pullback functor

$$QCoh(X_\Sigma/T) \longrightarrow QCoh(X_{\Sigma'}/T)$$

  is fully faithful.
  - This should be checkable by reducing to when Σ is a single cone and using the combinatorial model.

A first observation

Ntn Given a presentable stable ∞-category $\mathcal{C}$ and collection S of objects of $\mathcal{C}$, write $\langle S \rangle \subset \mathcal{C}$ for the closure of S under colimits and shifts

> Combining what we've shown so far, we see:

Prop Let (N, Σ) be a smooth projective fan. Then

(1) $\mathrm{im}(K_\Sigma) \subset \mathrm{Sh}(M_{\mathbb{R}})$ is closed under colimits and shifts.

(2) We have $\mathrm{im}(K_\Sigma) = \langle \omega_{m+\sigma^\vee} \mid m \in M, \sigma \in \Sigma \rangle$

(3) $F \in \mathrm{im}(K_\Sigma)$ if and only if for each $\sigma \in \Sigma$, we have

$$F * \omega_{\sigma^\vee} \in \langle \omega_{m+\sigma^\vee} \mid m \in M \rangle$$

(4) $\mathrm{im}(K_\Sigma)$ contains the unit for convolution and is closed under convolution products

Motivation for the Characterization

Ex. Let's consider the most simple example $N = \mathbb{Z}$ and $\sigma = \mathbb{R}_{\geq 0}$ to see why $\mathrm{Im}(\kappa_\sigma)$ is smaller than $\mathrm{Cons}_{P_\sigma}(M_{\mathbb{R}})$

- Of course, this example is affine, so it doesn't technically fit into the framework. But the simplest projective example with a problem is 2-dimensional, so let's understand this first

In this case, $\sigma^\vee = \mathbb{R}_{\geq 0}$ and $\sigma^\perp = 0$, so the FLTZ stratification is

As we saw last time, constructible sheaves are functors out of the poset

$$P_\sigma = \cdots \to \bullet \leftarrow \bullet \to \bullet \leftarrow \bullet \to \bullet \leftarrow \bullet \to \bullet \leftarrow \cdots$$

On the other hand, the combinatorial model tells us that

$QCoh(\mathbb{A}^1/\mathbb{G}_m)$ is presheaves on the poset

$$\Theta(\sigma) = \{m + \mathbb{R}_{\geq 0} \mid m \in \mathbb{Z}\}$$
$$\simeq \mathbb{Z}^{op}$$

One can show that the functor

$$K_\sigma : QCoh(\mathbb{A}^1/\mathbb{G}_m) \simeq Fun(\mathbb{Z}, Sp) \longrightarrow Fun(P_\sigma, Sp) \simeq Cons_{P_\sigma}(\mathbb{R})$$

is induced by restriction along the functor $P_\sigma \longrightarrow \mathbb{Z}$ that inverts all left-pointing arrows.

$$\cdots \to \bullet \xleftarrow{\sim} \bullet \to \bullet \xleftarrow{\sim} \bullet \to \bullet \xleftarrow{\sim} \bullet \to \bullet \xleftarrow{\sim} \cdots$$

$$\Big\downarrow \text{invert}$$

$$\cdots \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \cdots$$

Exercise We can't refine the FLTZ stratification to make $\mathbb{Z}$ the exit-path category.

> So if we want a characterization of $\mathrm{im}(\kappa)$ in terms of sheaf theory, we'll have to do something different

Ex. For the standard fan Σ in $\mathbb{Z}^2$ defining $\mathbb{P}^2$ (generated by e_1, e_2, and $-(e_1+e_2)$), one can also show that $\mathrm{im}(\kappa_\Sigma)$ is strictly smaller than $\mathrm{Cons}_{P_\Sigma}(\mathbb{R}^2)$. The picture is a bit involved to draw, but is well-explained on p. 70 of Bai-Hu

An overview of microsupport

What type of object. X (real analytic) manifold

$$\mathrm{Sh}(X, \mathrm{Sp}) \ni F \longmapsto \mu\mathrm{supp}(F) \subset T^\vee X$$

Idea $\mu\mathrm{supp}(F)$ records those codirections in which propagation of sections of F is **obstructed**

> i.e., the codirections in which F **fails** to be locally constant

Ex. Let's illustrate the idea in an example. Let $j : (a,b) \hookrightarrow \mathbb{R}$ be an open interval, and consider $\mathbb{1}_{(a,b)} = j_!(\mathbb{1})$.

> Since $\mathbb{1}_{(a,b)}$ is locally constant above (a,b), the microsupport above (a,b) is just be contained in the zero section

> Since $\mathbb{1}_{(a,b)}$ is zero on $\mathbb{R} \setminus [a,b]$, the microsupport above $\mathbb{R} \setminus [a,b]$ is empty.

> To calculate the microsupport above a, we pick a small interval (generally: ball) $(a-\varepsilon, a+\varepsilon)$ centered at a, and compare the 'past' to 'future' in a codirection $\xi \in T_a^\vee \mathbb{R} \simeq \mathbb{R}$

- If $\xi > 0$, this is comparing

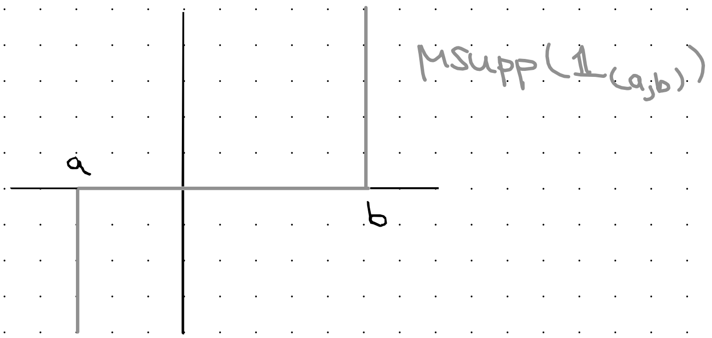
$$\begin{array}{ccc} \Gamma((a-\varepsilon, a+\varepsilon); \mathbb{1}_{(a,b)}) & \longrightarrow & \Gamma((a-\varepsilon, a); \mathbb{1}_{(a,b)}) \\ \| & & \| \\ 0 & & 0 \end{array}$$

same $\Rightarrow$ not in microsupport

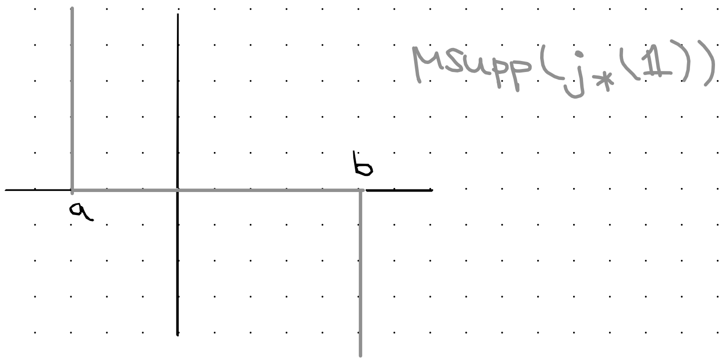
- If $\xi < 0$, this is comparing

$$\begin{array}{ccc} \Gamma((a-\varepsilon, a+\varepsilon); \mathbb{1}_{(a,b)}) & \longrightarrow & \Gamma((a, a+\varepsilon); \mathbb{1}_{(a,b)}) \\ \| & & \wr\| \\ 0 & & S \end{array}$$

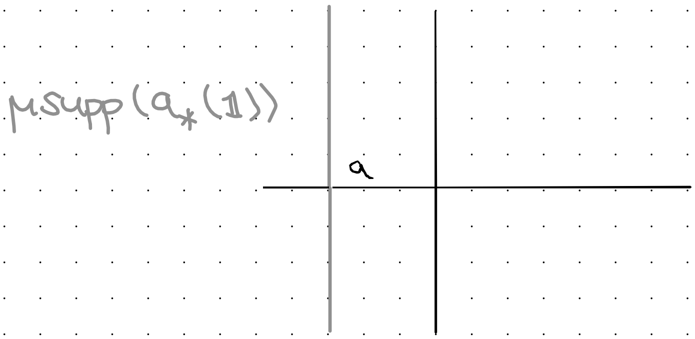
different $\Rightarrow$ in microsupport.



Dually. The microsupport of $j_*(\mathbb{1})$ is



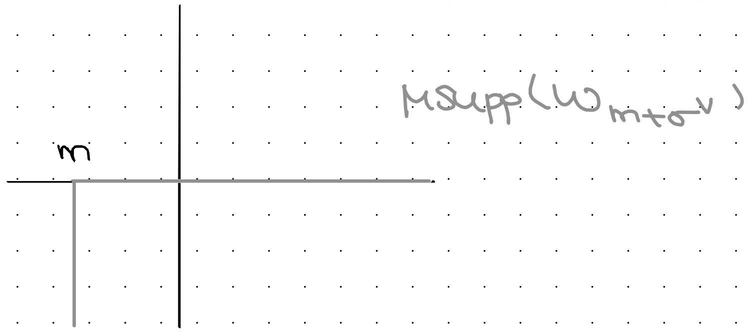
Ex. If $a: \{a\} \hookrightarrow \mathbb{R}$ is the inclusion of a point, then the microsupport of the skyscraper sheaf $a_*(\mathbb{1})$ is the cotangent fiber $T_a^\vee \mathbb{R} \simeq \mathbb{R}^\vee$.



Ex. Consider the lattice $\mathbb{Z}$ with cone $\sigma = \mathbb{R}_{\geq 0}$. Then each dualizing sheaf

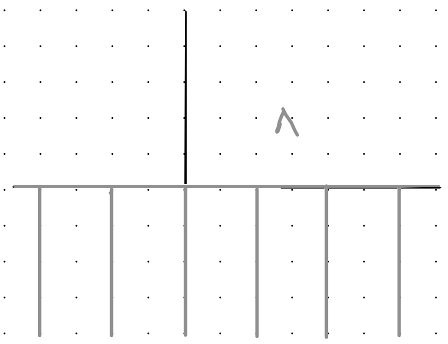
$$\omega_{m+\sigma^\vee} \simeq \mathbb{1}_{(m,\infty)}[1] \qquad \text{for } m \in \mathbb{Z}$$

has microsupport



In particular, each has microsupport in

$$\Lambda = (\text{zero section}) \cup \coprod_{m \in \mathbb{Z}} \{m\} \times \mathbb{R}_{\leq 0}.$$



In this case, it will turn out that

$$\operatorname{im}(\kappa : QCoh(\mathbb{A}^1/\mathbb{G}_m) \longrightarrow Sh(\mathbb{R}))$$

is the subcategory of sheaves with microsupport in Λ.

Properties of microsupport

Properties $\mu supp(F) \subset T^\vee X$ is / satisfies

(1) Closed

(2) $\mathbb{R}_{>0}$ - conic

(3) Coisotropic (a hard theorem of Kashiwara - Schapira)

(4) $\mu supp(F) \cap (\text{zero section}) = \pi(\mu supp(F)) = Supp(F)$

$\pi : T^\vee X \longrightarrow X$ projection.

(5) $\mu supp(\mathbb{D}(F)) = -\mu supp(F)$ where $\mathbb{D}$ denotes Verdier duality

Categorical properties (clear from the definition)

(0) $\mu supp(0) = \emptyset$

(1) Given a fiber sequence $F' \longrightarrow F \longrightarrow F''$, we have

$$\mu supp(F) \subset \mu supp(F') \cup \mu supp(F'')$$

$$\mu supp(F[n]) = \mu supp(F).$$

(2) If F' is a retract of F, then $\mu supp(F') \subset \mu supp(F)$.

Microsupport estimates (Guillermou–Viterbo). If $F_\bullet : A \longrightarrow Sh(X)$ is a diagram of sheaves on X, then

$$\mu supp\left(\operatorname*{colim}_{\alpha\in A} F_\alpha\right) \subset \overline{\bigcup_{\alpha\in A} \mu supp(F_\alpha)}$$

$$\mu supp\left(\lim_{\alpha\in A} F_\alpha\right) \subset \overline{\bigcup_{\alpha\in A} \mu supp(F_\alpha)}$$

Ntn For a closed $\mathbb{R}_{>0}$-conic subset $\Lambda \subset T^\vee X$ (usually taken to be subanalytic Lagrangian), write $Sh_\Lambda(X) \subset Sh(X)$ for the full subcategory spanned by those sheaves with microsupport contained in Λ.

Cor The subcategory $Sh_\Lambda(X) \subset Sh(X)$ is closed under limits and colimits. Hence $Sh_\Lambda(X)$ is presentable and the inclusion admits both a left and a right adjoint.

Microlocal characterization of constructibility

Thm. Let X be a real analytic manifold and let $X \to P$ be a stratification of X by subanalytic subsets (we always assume such stratifications are locally finite.) TFAE for $F \in Sh(X)$

(1) F is P-constructible.

(2) $\mu supp(F) \subset \bigcup_{p \in P} T^\vee_{X_p} X$, where $T^\vee_{X_p} X$ is the conormal bundle of X_p

Thm. TFAE for a sheaf F on X:

(1) F is constructible wrt **some** subanalytic stratification

(2) $\mu supp(F) \subset T^\vee X$ is a subanalytic Lagrangian.

(3) $\mu supp(F)$ is **contained** in a subanalytic Lagrangian.

Microlocal characterization of $\mathrm{im}(K_\Sigma)$

> The following microsupport estimate isn't difficult, but is key

Lem. Let (N, Σ) be a fan, and for $m \in M$, consider the dualizing sheaf $\omega_{m+\sigma^\vee} \in \mathrm{Cons}_{P_\Sigma}(M_{\mathbb{R}})$. Then

$$\mu\mathrm{supp}(\omega_{m+\sigma^\vee}) \subset \bigcup_{\tau \subset \sigma} (m + \tau^\perp) \times (-\tau)$$

inside $T^\vee M_{\mathbb{R}} \simeq M_{\mathbb{R}} \times N_{\mathbb{R}}$.

Def. Let (N, Σ) be a fan. The **FLTZ skeleton** $\Lambda_\Sigma \subset T^\vee M_{\mathbb{R}}$ is the $\mathbb{R}_{>0}$-conic Lagrangian

$$\Lambda_\Sigma = \bigcup_{m \in M, \sigma \in \Sigma} (m + \sigma^\perp) \times (-\sigma) \subset M_{\mathbb{R}} \times N_{\mathbb{R}}.$$

Cor. For any fan (N, Σ), we have

$$\mathrm{im}(K_\Sigma) \subset Sh_{\Lambda_\Sigma}(M_{\mathbb{R}}).$$

Toric mirror symmetry. If (N, Σ) is a smooth projective fan, then

$$K_\Sigma : QCoh(X_\Sigma / T) \longrightarrow Sh_{\Lambda_\Sigma}(M_{\mathbb{R}})$$

is an equivalence

> Next time, we'll given an overview of the proof.